JASWANTH MADA

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SUMMARY

Applied AI / Machine Learning Engineer specializing in multi-agent orchestration, production RAG pipelines, and LLM application development.

EDUCATION

Purdue University Northwest **Aug 2023 - May 2025**
Masters, Computer Science *Hammond, IN*

TECHNICAL SKILLS

- **Programming & Core ML**: Python, NumPy, Pandas, Scikit Learn, PyTorch
- **Agentic AI & Orchestration**: Crew AI, Multi Agent Systems, ReAct Pattern, Planning Pattern, Tool Use, Agent Memory Management, Guardrails & Output Validation, Human-In-The-Loop, Agent Callbacks
- **Cloud & AWS**: EC2, S3, Lambda, API Gateway, SQS, DynamoDB, ElasticCache(Redis), CloudWatch
- **Databases**: MySQL, PostgreSQL, DynamoDB, Redis
- **Software Engineering**: OOP, Data Structures & Algorithms, Design Patterns, Multithreading, Concurrency
- **LLMs & Retrieval Systems**: RAG, LangChain, Pinecone, Prompt Engineering, Embedding Based Search

EXPERIENCE

Accessifiers **Aug 2025 - Jun 2026**
Machine Learning Engineer

- Engineered an internal LLM-powered document Q&A system using Python, LangChain, and Pinecone, reducing time spent on policy and technical document lookups by 35% for research and engineering users.
- Implemented a scalable document ingestion and chunking pipeline on AWS (S3, Lambda) to process and refresh 100K+ pages of PDFs and markdown content with consistent retrieval behavior.
- Optimized prompt templates, retrieval depth, and citation logic using OpenAI APIs and prompt versioning, cutting irrelevant or partially grounded responses by 25% in internal validation.
- Automated testing and deployment workflows with GitHub Actions and pytest, validating prompt changes and retrieval outputs and lowering release-related issues by 30%.
- Instrumented latency, token usage, and error metrics using CloudWatch and logging, keeping average response time under 3 seconds while managing inference cost during peak usage.

PROJECTS

Autonomous Multi-Agent Job Application System | Python, LangGraph, MCP, Playwright, OpenRouter, Apify Github

- **Engineered a hierarchical multi-agent framework** using LangGraph to automate end-to-end job applications, implementing a sequential Manager-Worker state machine with deterministic fallback for CAPTCHAs and scam detection.
- **Reduced LLM token overhead** by implementing an active memory-management pipeline that prunes heavy DOM payloads from context windows beyond 18 conversation turns while preserving critical action history.
- Integrated **MCP-based browser and Gmail tooling** with asynchronous session management, dynamic tool discovery, timeout handling, graceful server recovery, and human-in-the-loop interaction for unknown application questions.
- **Built a resilient tool-execution runtime** with human-in-the-loop approval gates for high-risk actions, automatic exception recovery, and asynchronous MCP tool resolution.

Agentic Content Writing System

- **Built a production-grade Multi-Agent content writing system**, evolving from a single-agent baseline to a fully orchestrated, multi-role crew applying all **core agentic AI patterns** in a unified implementation.
- **Implemented the ReAct (Reasoning + Acting) pattern**, enabling agents to **reason step-by-step**, dynamically select tools, and **self-correct** before producing outputs, reducing hallucinations and improving task reliability.
- **Designed a multi-agent crew with specialized roles (Researcher, Writer, and Editor)** using **hierarchical and sequential process modes** with inter-agent task delegation and **context passing** between crew members.
- **Implemented guardrails and validation layers** to intercept malformed outputs, trigger **retry logic**, and enforce editorial constraints like **word count bounds** before output propagation.
- **Managed short-term, long-term, entity, and contextual memory** across agent interactions, enabling the crew to **retain context across tasks** and reduce redundant information retrieval.

Achievements

- **ICPC:**Ranked 43 out of 89 teams in the USA Mid-Center Regional ICPC Contest.
- **IEEE Xtreme:**Ranked 29 out of 149 teams in the 18th Annual Global IEEE Xtreme Programming Competition.